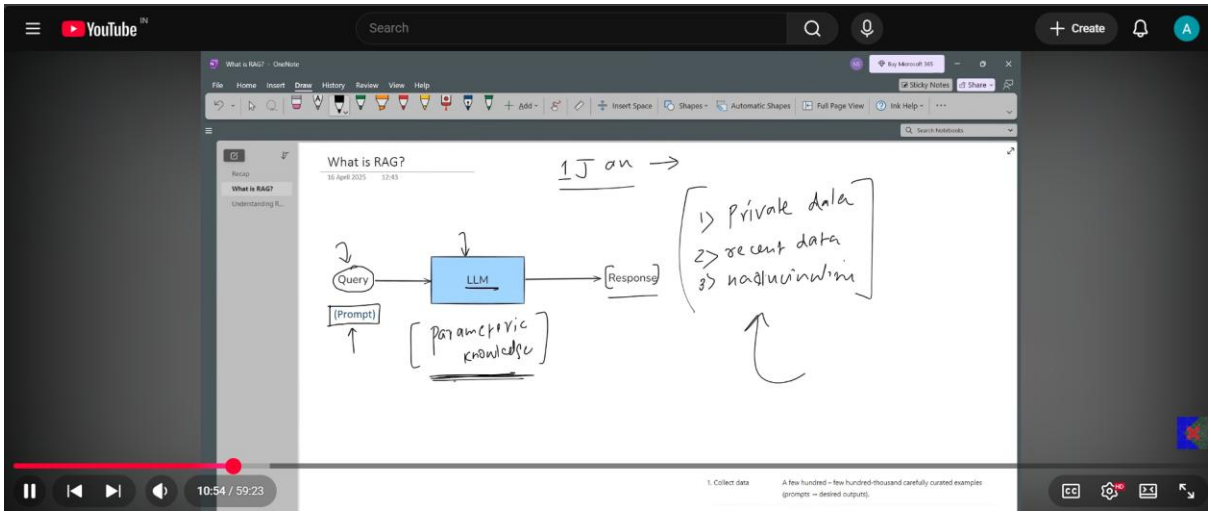


fine-tuning

pretrained LLM

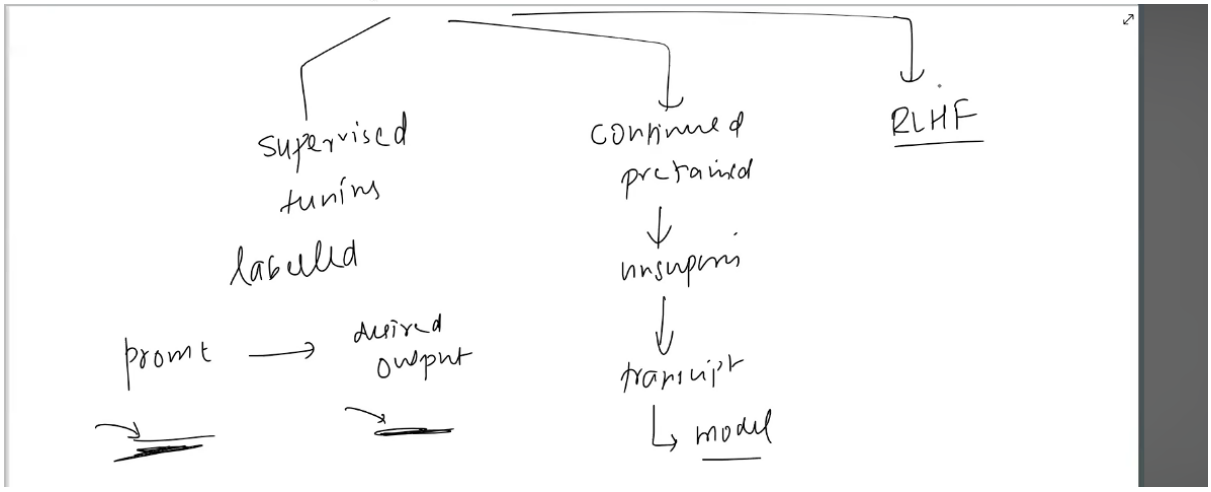
small → domain specific

student – LLM

ends → pretrain

train → fine tuning

1. Collect data	A few hundred – few hundred-thousand carefully curated examples (prompts → desired outputs).
2. Choose a method	Full-parameter FT, LoRA/QLoRA, or parameter-efficient adapters.
3. Train for a few epochs	You keep the base weights frozen or partially frozen and update only a small subset (LoRA) or all weights (full FT).
4. Evaluate & safety-test	Measure exact-match, factuality, and hallucination rate against held-out data; red-team for safety.



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In-Context Learning is a core capability of Large Language Models (LLMs) like GPT-3/4, Claude, and Llama, where the model learns to solve a task purely by seeing examples in the prompt—without updating its weights.

```
Below are exa
Use these exa

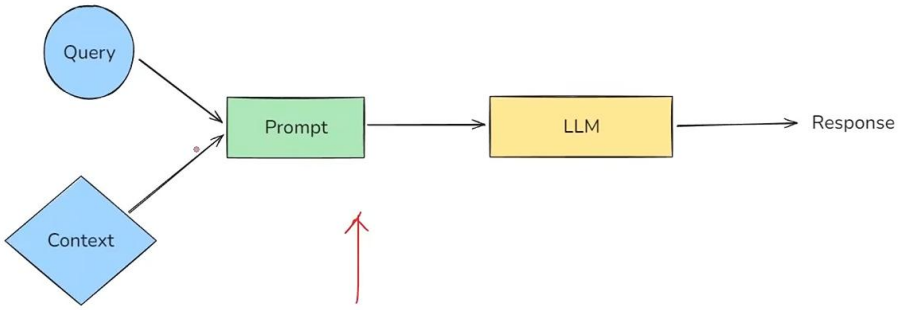
Text: I love
```

In-Context Learning is a core capability of Large Language Models (LLMs) like GPT-3/4, Claude, and Llama, where the model learns to solve a task purely by seeing examples in the prompt—without updating its weights.

LLM → emergent property

An **emergent property** is a behaviour or ability that suddenly appears in a system when it reaches a certain scale or complexity—even though it was not explicitly programmed or expected from the individual components.

RAG is a way to make a language model (like ChatGPT) smarter by giving it extra information at the time you ask your question.




```
"""You are a helpful assistant.
Answer the question ONLY from the provided context.
If the context is insufficient, just say you don't know.

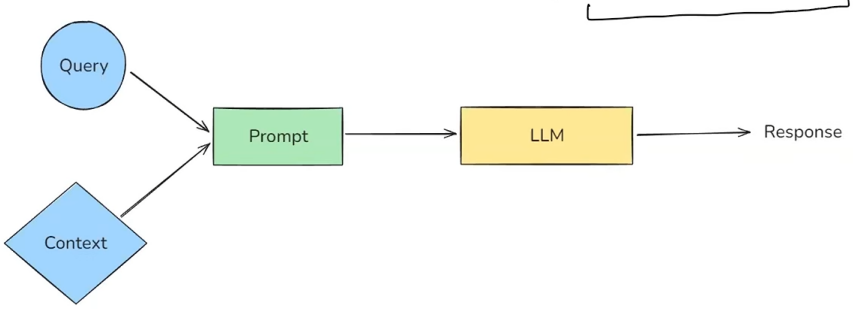
{context}
Question: {question}"""
```

Search Notebooks

Understanding RAG

16 April 2025 12:44

RAG → Information retrieval + text generation



- Indexing
- Retrieval
- Augmentation
- Generation

Indexing - **Indexing** is the process of **preparing your knowledge base so that it can be efficiently searched** at query time. This steps consists of 4 sub-steps.

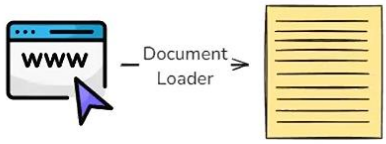
1. Document Ingestion - You load your source knowledge into memory.

   Examples:

   - PDF reports, Word documents
   - YouTube transcripts, blog pages
   - GitHub repos, internal wikis
   - SQL records, scraped webpages

   Tools:

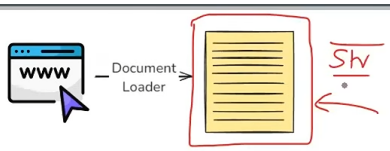
   - LangChain loaders (`PyPDFLoader` , `YoutubeLoader` , `WebBaseLoader` , `GitLoader` , etc.)



2. Text Chunking - Break large documents into small, semantically meaningful **chunks**

   Why chunk?

   - LLMs have context limits (e.g., 4K–32K tokens)



LLM → Context

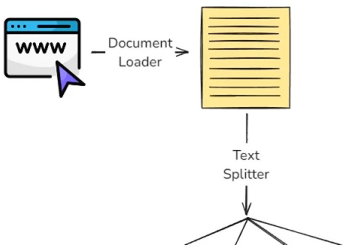
2. Text Chunking - Break large documents into small, semantically meaningful **chunks**

   Why chunk?

   - LLMs have context limits (e.g., 4K–32K tokens)
   - Smaller chunks are more focused → better semantic search

   Tools:

   - `RecursiveCharacterTextSplitter` , `MarkdownHeaderTextSplitter` , `SemanticChunker`



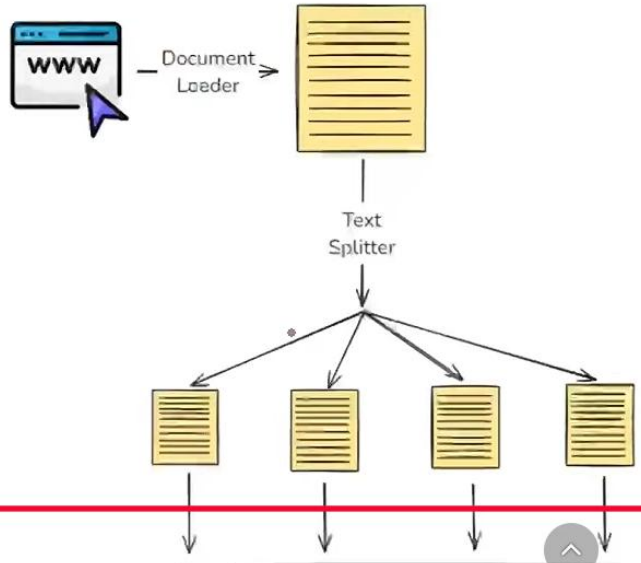
3. Embedding Generation - Convert each chunk into a **dense vector** (embedding) that captures its meaning.

🔍 Why embeddings?

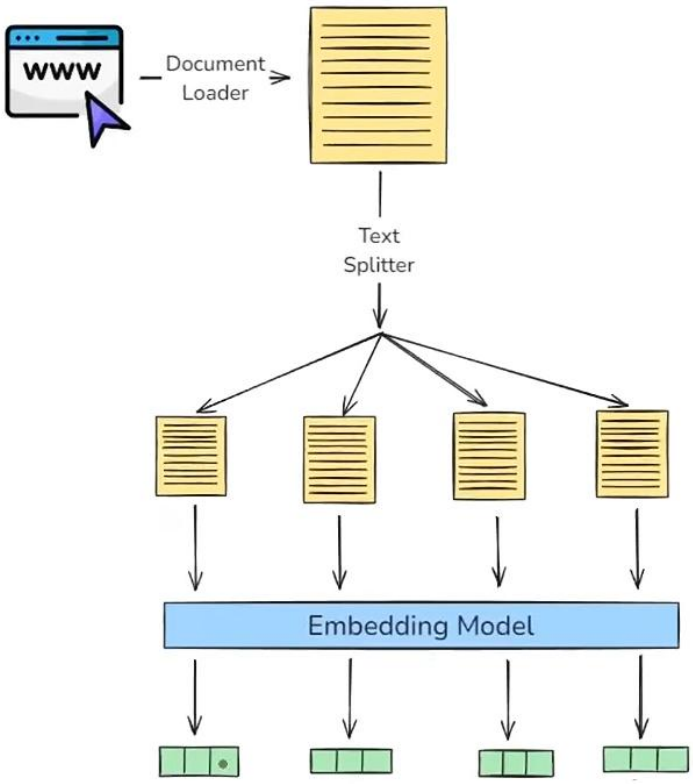
- Similar ideas land close together in vector space
- Allows fast, fuzzy semantic search

🔧 Tools:

- `OpenAIEmbeddings`, `SentenceTransformerEmbeddings`, `InstructorEmbeddings`, etc.



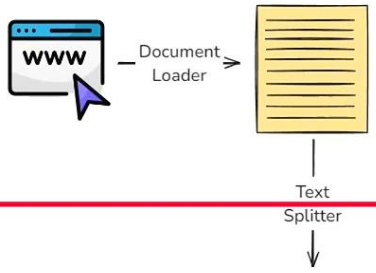
- `OpenAIEmbeddings`, `SentenceTransformerEmbeddings`, `InstructorEmbeddings`, etc.



4. Storage in a Vector Store - Store the vectors along with the original chunk text + metadata in a **vector database**.

   Vector DB options:

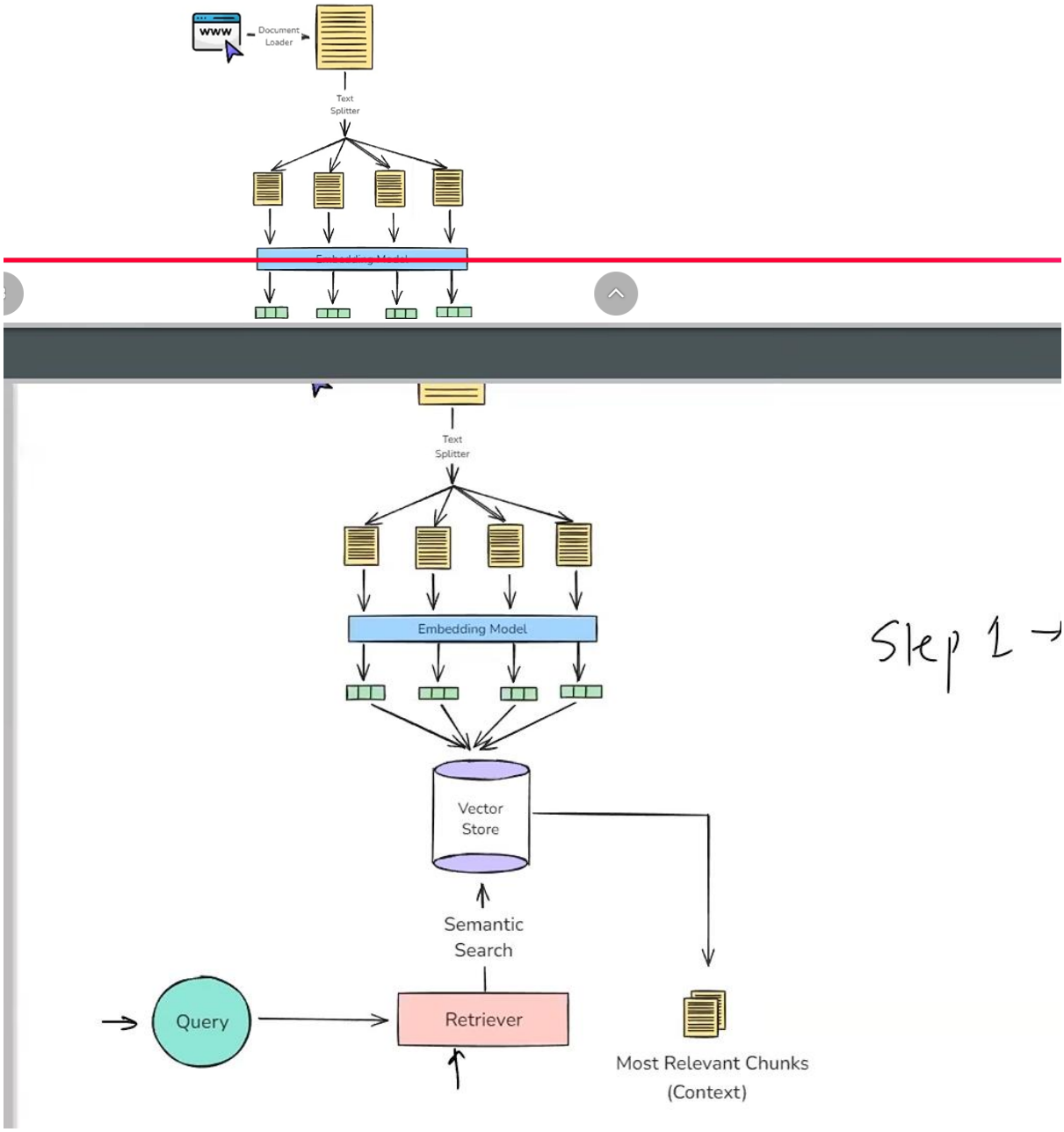
   - Local: `FAISS`, `Chroma`
   - Cloud: `Pinecone`, `Weaviate`, `Milvus`, `Qdrant`



Retrieval - **Retrieval** is the *real-time* process of **finding the most relevant pieces of information** from a pre-built index (created during indexing) based on the user's question.

It's like asking:

> "From all the knowledge I have, which 3–5 chunks are most helpful to answer this query?"



Generation - **Generation** is the final step where a **Large Language Model (LLM)** uses the **user's query** and the **retrieved & augmented context** to generate a response.

